

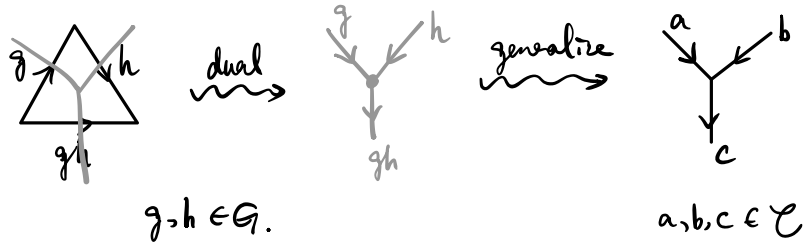
Fusion Categories and Turaev-Viro-Levin-Wen Model

spacetime space

Motivation.

Generalize TQDM of a group G .

Hilbert space:

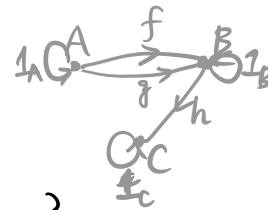


$\mathcal{C} \begin{cases} \text{types: } a, b, c, \dots \\ \text{multiplication / tensor product } a \times b \end{cases}$

3.1. Categories

Def. A category $\mathcal{C}$ consists of

- A collection of objects $\text{Obj}(\mathcal{C}) = \{A, B, \dots\}$
- A collection of morphisms $\text{Hom}(A, B)$ for $\forall A, B \in \text{Obj}(\mathcal{C})$
- A composition map for $\forall A, B, C \in \text{Obj}(\mathcal{C}) = \mathcal{C}$

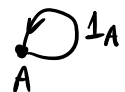


$$C_{A,B,C} : \text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$$

$$(f, g) \mapsto g \circ f$$

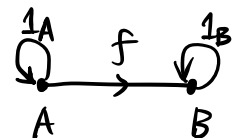
$$\begin{array}{c} \bullet \xrightarrow{f} \bullet \xrightarrow{g} \bullet \\ A \quad B \quad C \end{array} \mapsto \begin{array}{c} \bullet \xrightarrow{g \circ f} \bullet \\ A \quad C \end{array}$$

- An identity morphism $1_A = \text{id}_A \in \text{Hom}(A, A)$



such that

- $(f \circ g) \circ h = f \circ (g \circ h)$ associativity
- $f \circ 1_A = f = 1_B \circ f$ identity



Examples

(1) Set :

$$\text{set } A \xrightarrow{\text{map } f} \text{set } B$$

(2) a group G :



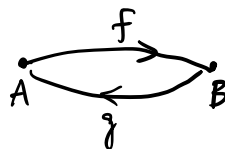
$$\text{Obj}(\mathcal{G}) = \{*\}$$

$$\text{Hom}(*, *) = G$$

$$* \xrightarrow{g} * \xrightarrow{h} * = * \xrightarrow{g \cdot h} *$$

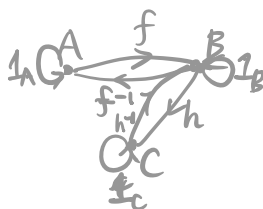
$$1_* = e \in G$$

Def: An isomorphism $f: A \rightarrow B$



$$\begin{cases} g \circ f = 1_A \\ f \circ g = 1_B \end{cases}$$

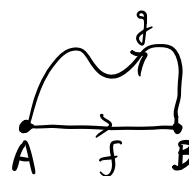
(3) groupoid : A cat $\mathcal{G}$ is a groupoid if every morphism is isomorphism.



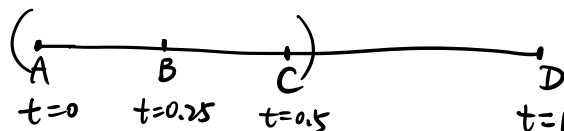
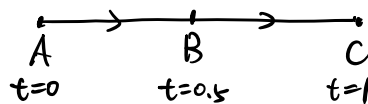
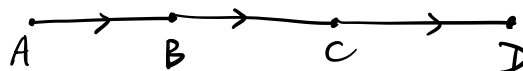
(4) fundamental groupoid of a topological space M :

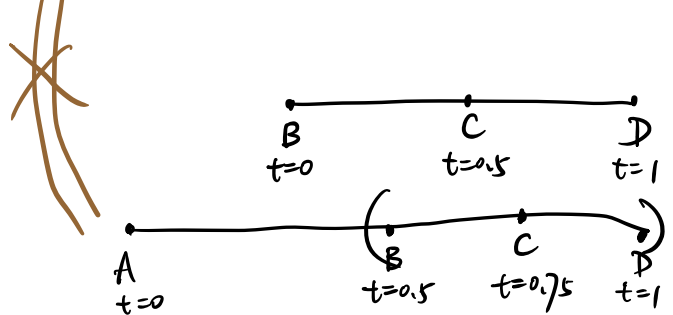
$$\text{Obj} = \{\text{points in } M\}$$

$$\text{Hom}(A, B) = \{\text{Paths from } A \text{ to } B\} / \text{homotopy equivalence}$$



$$\text{path: } I = [0, 1] \rightarrow M$$





$$(5) \text{ Grp : } \text{group } G \xrightarrow[\text{homomorphisms}]{\text{group}} \text{group } G'$$

$$(6) \text{ Vec : } \text{vector space } V \xrightarrow[\text{maps}]{\text{linear}} \text{vector space } W$$

$$(7) \text{ Top : } \text{topological space } M \xrightarrow[\text{maps}]{\text{continuous}} \text{topological space } N$$

Def A functor from categories $\mathcal{C}$ to $\mathcal{D}$ is a map sending

- any object $X \in \mathcal{C}$ to an object $F(X)$ in $\mathcal{D}$.
- any morphism $f: X \rightarrow Y$ in $\mathcal{C}$ to a morphism $F(f): F(X) \rightarrow F(Y)$ in $\mathcal{D}$.

such that

- F preserves identity : $F(1_X) = 1_{F(X)}$
- F preserves composition : $F(g \circ f) = F(g) \circ F(f)$

$$\begin{array}{ccc} \begin{array}{c} \bullet \xrightarrow{f} \bullet \\ X \quad Y \\ \mathcal{C} \end{array} & \xrightarrow{F} & \begin{array}{c} \bullet \xrightarrow{F(f)} \bullet \\ F(X) \quad F(Y) \\ \mathcal{D} \end{array} \end{array}$$

Examples. (1) $\begin{array}{ccc} * \curvearrowright g \in G & \xrightarrow{F} & V \curvearrowright \rho(g) \\ G & \xrightarrow{F} & \text{Vec} \end{array}$

representation of G .

$$(2) H_n : \text{Top} \longrightarrow \text{Abel}$$

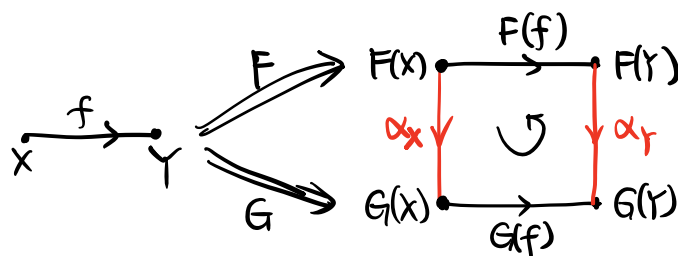
$$\begin{array}{ccc} M & & H_n(M) \\ f \downarrow & \xRightarrow{F} & \downarrow f_* \\ N & & H_n(N) \end{array}$$

Def. Given two functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, a natural transformation

$\alpha: F \Rightarrow G$ assigns to every object X in $\mathcal{C}$ a morphism

$\alpha_X: F(X) \rightarrow G(X)$ in $\mathcal{D}$, s.t.

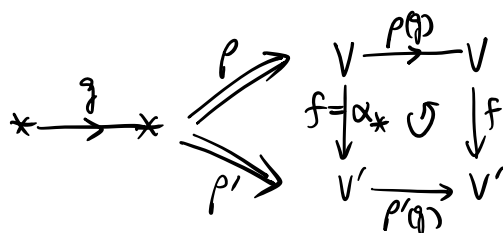
$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \downarrow \alpha_X & & \downarrow \alpha_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$



Example. a group G , given two rep (functors) $\rho: G \rightarrow V$
 $\rho': G \rightarrow V'$

a natural transformation (intertwiner) is a map $f: V \rightarrow V'$,

s.t. $f \circ \rho(g) = \rho'(g) \circ f$ for $g \in G$.



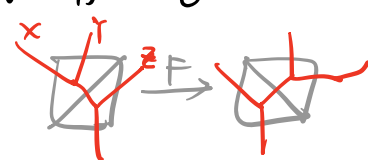
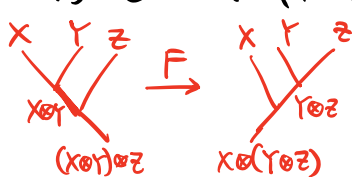
3.2. Fusion categories

Def. A monoidal category consists of

- a category $\mathcal{C}$
- a tensor product functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$
- a unit object $1 \in \mathcal{C}$
- a natural isomorphism

$$a_{X,Y,Z}: (X \otimes Y) \otimes Z \rightarrow X \otimes (Y \otimes Z), \quad \forall X, Y, Z \in \mathcal{C}$$

$\uparrow$
associator
F move



- natural isomorphism (left/right unitors) for $X \in \mathcal{C}$

$$l_X: 1 \otimes X \rightarrow X$$

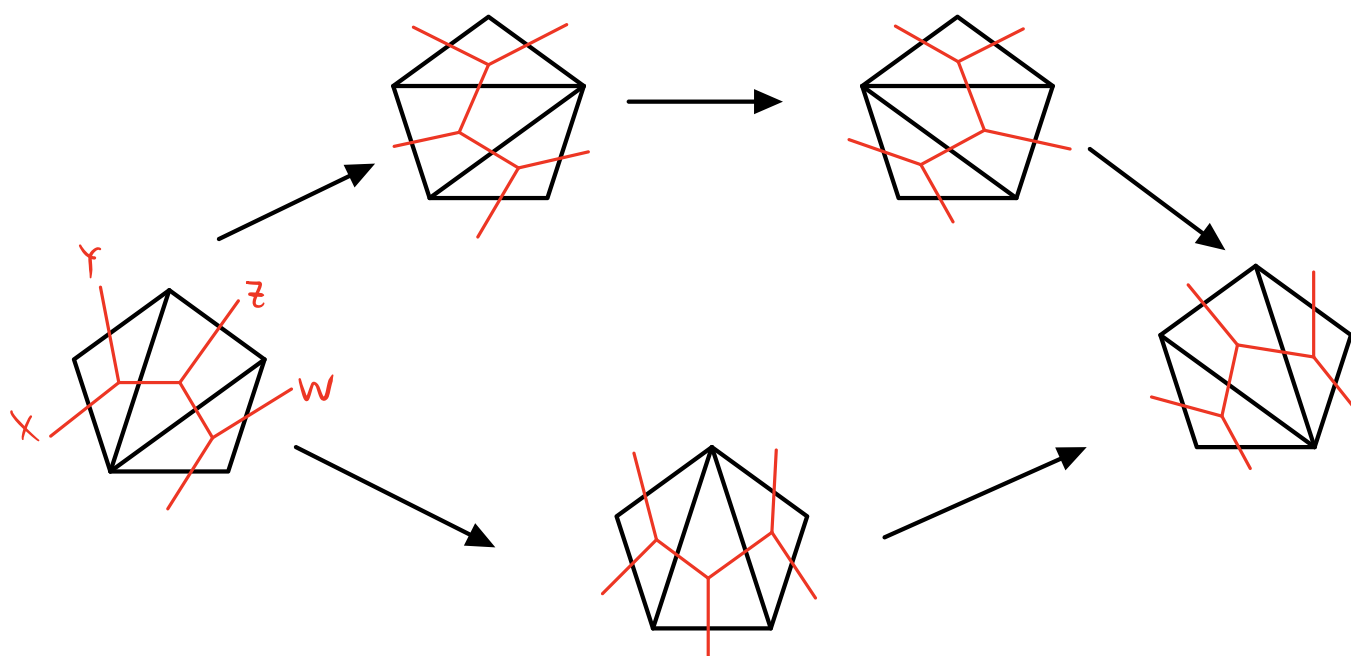
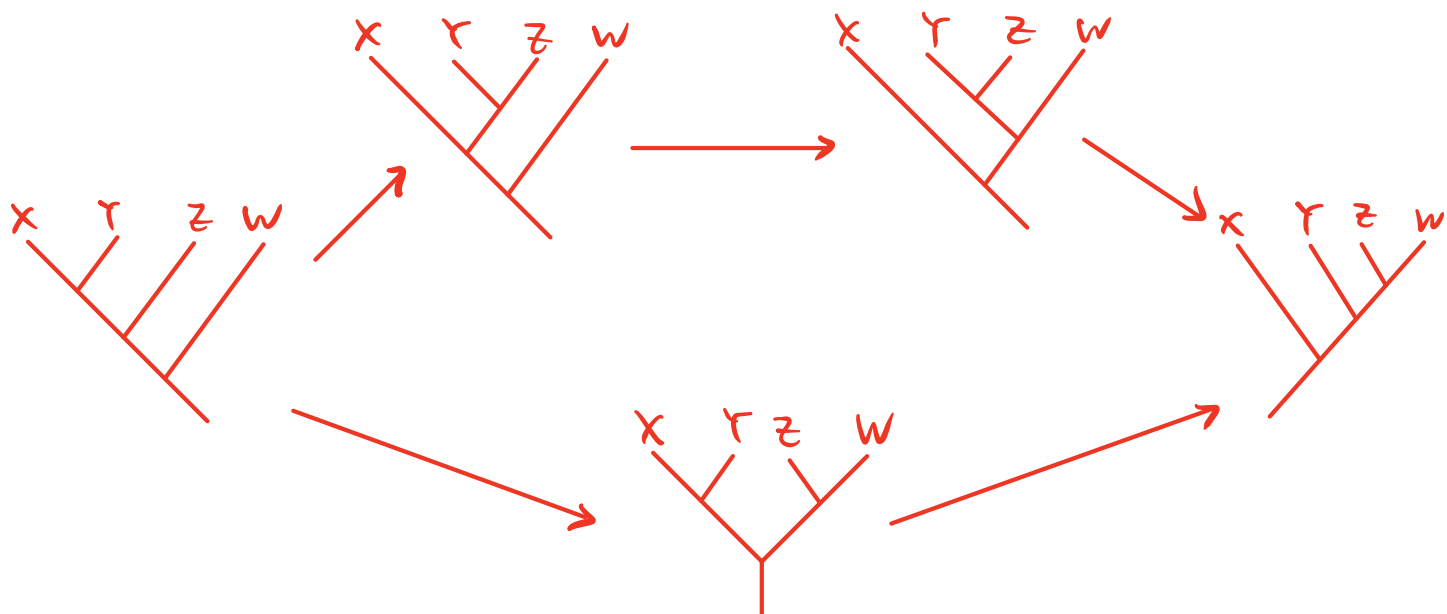
$$r_X: X \otimes 1 \rightarrow X$$

such that

$$\begin{array}{ccc} (X \otimes 1) \otimes Y & \xrightarrow{a_{X,1,Y}} & X \otimes (1 \otimes Y) \\ \downarrow r_X \otimes 1_Y & \searrow \quad \quad \quad \swarrow 1_X \otimes r_Y & \\ & X \otimes Y & \end{array}$$

- pentagon equation:

$$\begin{array}{ccccc} & & & a_{X,Y \otimes Z,W} & \\ & & & \nearrow & \\ & & (X \otimes (Y \otimes Z)) \otimes W & \xrightarrow{\quad} & X \otimes ((Y \otimes Z) \otimes W) \\ a_{X,Y,Z} \otimes 1_W \nearrow & & & & \searrow 1_X \otimes a_{Y,Z,W} \\ (X \otimes Y) \otimes Z \otimes W & & & & X \otimes (Y \otimes (Z \otimes W)) \\ & \searrow a_{X \otimes Y, Z \otimes W} & & \nearrow a_{X,Y,Z \otimes W} & \\ & (X \otimes Y) \otimes (Z \otimes W) & & & \end{array}$$



Examples

(1) $\text{Rep}(G)$.

$\text{obj} = \text{vector space}$

$\text{Hom} = G\text{-invariant linear maps.}$

$$V \otimes W$$

$1 = \text{trivial rep on } \mathbb{C}.$

$\alpha_{x,y,z}$ is trivial

(2) Vec_G .

$\text{obj} = V_g \text{ for } g \in G$

$$\text{Hom}(V_g, V_h) = \begin{cases} \mathbb{C}, & g = h \\ 0, & g \neq h \end{cases}$$

$$V_g \otimes V_h = V_{gh}$$

α_{V_g, V_h, V_k} is trivial

$1 = V_e$, e identity element in G .

(3) $\text{Vec}_G^{\nu_3}$:

$$\alpha_{V_g, V_h, V_k} := \nu_3(g, h, k) \in U(1)$$

$$\text{pentagon eq} \Leftrightarrow d\nu_3 = 1$$

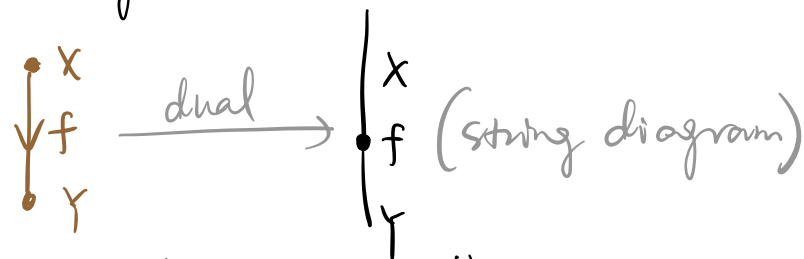
$$\Leftrightarrow \nu_3 \in Z^3(G, U(1))$$

Def.

A fusion category is a rigid, semisimple, linear monoidal category with finitely many isomorphism classes of simple objects such that $\text{Hom}(1, 1) = \mathbb{C}$.

with dual $a \oplus b$ hom set is vector space.
↑ ↑ ↑

3.3. String diagram

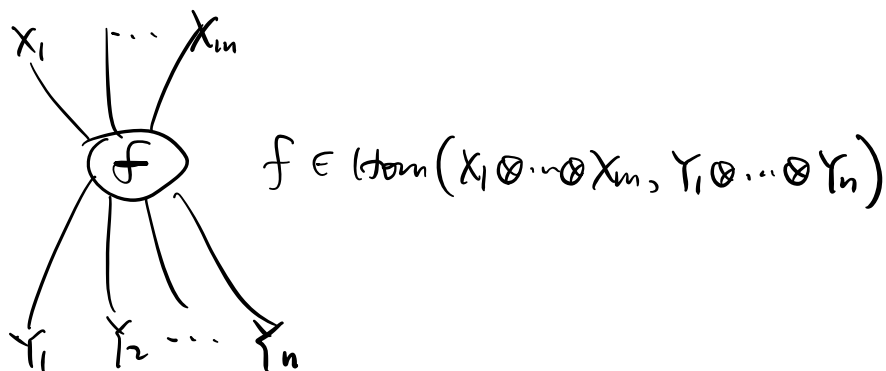
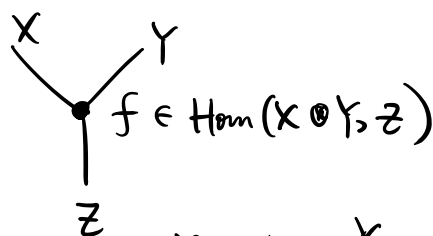


$$\begin{array}{c} X \\ | \\ \bullet \\ | \\ X \end{array} \text{id}_X = \begin{array}{c} X \\ | \\ X \end{array}, \quad \begin{array}{c} X \\ | \\ \bullet \\ | \\ Y \\ | \\ \bullet \\ | \\ Z \end{array} = \begin{array}{c} X \\ | \\ \bullet \\ | \\ Z \end{array} \circ \begin{array}{c} Y \\ | \\ \bullet \\ | \\ Z \end{array}$$

tensor: $X \otimes Y$

$$\begin{array}{c} | \\ \otimes \\ | \\ X \quad Y \end{array} \quad \begin{array}{c} X \\ | \\ \bullet \\ | \\ Y \end{array} \quad \begin{array}{c} X' \\ | \\ \bullet \\ | \\ Y' \end{array} = \begin{array}{c} X \otimes X' \\ | \\ \bullet \\ | \\ Y \otimes Y' \end{array} \text{ } f \otimes f'$$

$$\begin{array}{c} f \\ | \\ \bullet \\ | \end{array} \quad \begin{array}{c} | \\ g \\ | \\ \bullet \\ | \end{array} = \begin{array}{c} | \\ f \\ | \\ \bullet \\ | \\ g \\ | \\ \bullet \\ | \end{array} = \begin{array}{c} | \\ f \\ | \\ \bullet \\ | \\ g \end{array}$$

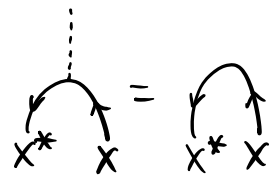


identity 1: $\begin{array}{c} \vdots \\ | \\ \bullet \\ | \\ \vdots \end{array} =$

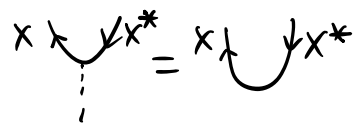
$$\begin{array}{c} X \\ | \\ \bullet \\ | \\ \vdots \end{array} = \begin{array}{c} X \\ | \\ \bullet \\ | \\ f \end{array} \quad f \in \text{Hom}(X, 1)$$

$$\begin{array}{c} \vdots \\ | \\ \bullet \\ | \\ Y \end{array} = \begin{array}{c} \bullet \\ | \\ f \\ | \\ Y \end{array} \quad f \in \text{Hom}(1, Y)$$

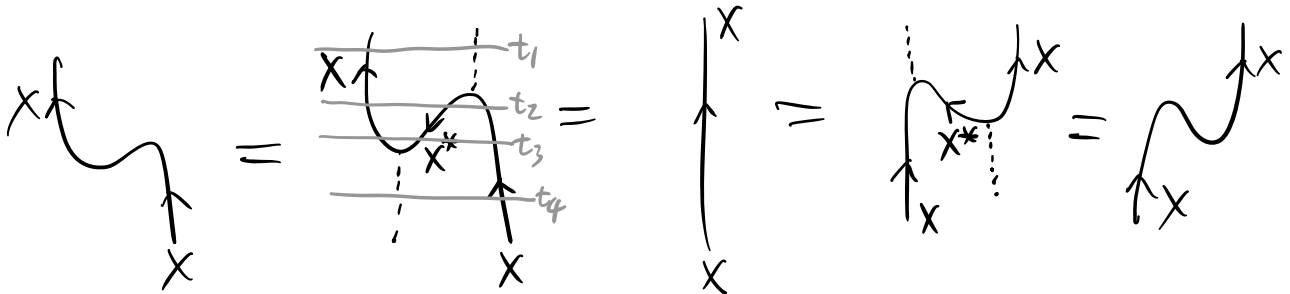
duals: unit $i_X: 1 \rightarrow X^* \otimes X$



counit $e_X: X \otimes X^* \rightarrow 1$

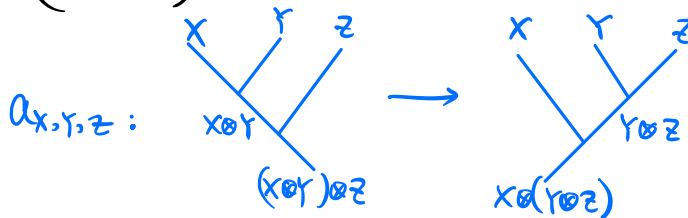


Sit.



$$\begin{array}{ccccccc}
 X \otimes 1 & \xrightarrow{id_X \otimes i_X} & X \otimes (X^* \otimes X) & \xrightarrow{a_{X, X^*, X}^{-1}} & (X \otimes X^*) \otimes X & \xrightarrow{e_X \otimes id_X} & 1 \otimes X \\
 t_1 & & t_2 & & t_3 & & t_4 \\
 & \searrow \gamma_X & & \searrow & & \searrow \ell_X & \\
 & & X & & & &
 \end{array}$$

associator (F move):



In simple object basis:

$$V_c^{ab} := \text{Hom}(a \otimes b, c) \ni \begin{array}{c} a \quad b \\ \diagdown \quad \diagup \\ \mu \\ | \\ c \end{array} \leftrightarrow |a, b; c, \mu\rangle$$

$$N_c^{ab} := \dim V_c^{ab}$$

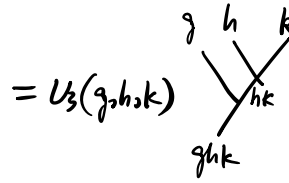
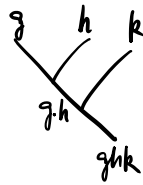
$$\begin{array}{c} a \quad b \quad c \\ \diagdown \quad \diagup \quad \diagup \\ \mu \quad e \quad \nu \\ | \\ d \end{array} = \sum_{f, \alpha, \beta} \left(F_{d, f, \alpha, \beta}^{abc} \right) \begin{array}{c} a \quad b \quad c \\ \diagdown \quad \diagup \quad \diagup \\ \mu \quad \alpha \quad \beta \\ | \\ d \end{array}$$

parameters
row col

$$\sum_e N_e^{ab} N_d^{ec} = \sum_f N_f^{bc} N_d^{af}$$

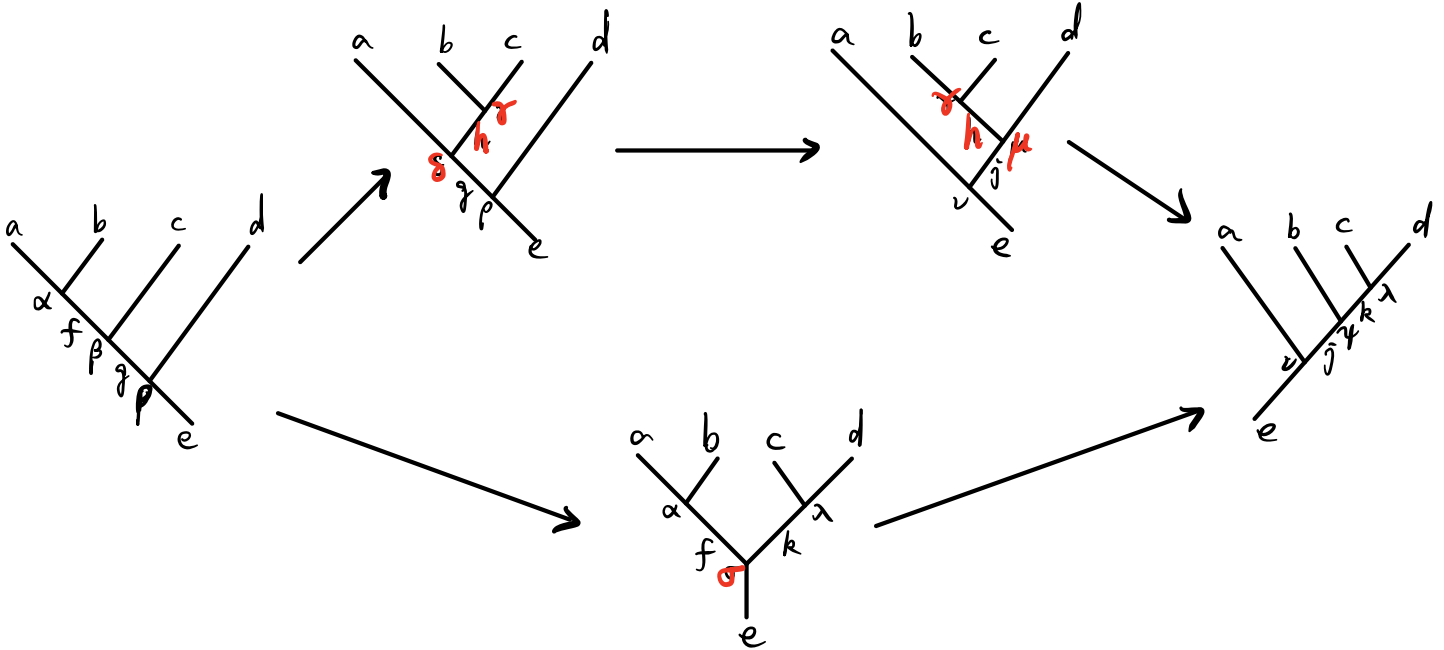
unitary fusion category: F is unitary.

$$\mathcal{C} = \text{Vec}_G^{\mathcal{U}_3} :$$



$$\mathcal{U}_3(g, h, k) := \left(F_{ghk}^{g, h, k} \right)_{1,1}$$

pentagon eq:



$$\sum_{h, \gamma, \delta, \mu} (F_g^{abc})_{f\alpha\beta, h\gamma\delta} (F_e^{ahd})_{g\delta\mu, j\mu\nu} (F_j^{bcd})_{h\gamma\mu, k\lambda\gamma}$$

$$= \sum_{\sigma} (F_e^{fed})_{g\beta\mu, k\lambda\sigma} (F_e^{abk})_{f\alpha\sigma, j\gamma\nu}$$

important relation:

$$\sum_{k, \mu} \begin{array}{c} \text{diagram} \\ \text{diagram} \end{array} = \begin{array}{c} \text{diagram} \\ \text{diagram} \end{array}$$

$$\sum_{k, \mu} |ij; k, \mu\rangle \langle ij; k, \mu| = I$$

$$\begin{array}{c} |k' \\ \text{---} \mu' \\ \bigcirc \\ \mu \\ \text{---} k \end{array} \hat{=} \delta_{kk'} \delta_{\mu\mu'} \quad \Bigg| \quad k$$

$$\langle i; k' \mu' | i; k \mu \rangle = \delta_{kk'} \delta_{\mu\mu'}$$

quantum dimension:

d_X of a simple object $X \in \mathcal{C}$ is

$$d_X := \bigcirc \nearrow X = X^* \begin{array}{c} 1 \\ \vdots \\ \bigcirc \\ \vdots \\ 1 \end{array} \nearrow X \in \text{Hom}(1, 1) = \mathbb{C}$$

$$d_X > 0.$$

property: $d_i d_j = \sum_k N_k^{ij} d_k$

proof: $d_i d_j = \bigcirc \nearrow i \nearrow j = \sum_{k, \mu} \bigcirc \begin{array}{c} i \\ \mu \\ \bigcirc \\ \mu \\ j \end{array} \nearrow k$

$$= \sum_{k, \mu} \hat{j} \begin{array}{c} \mu \\ \bigcirc \\ \mu \end{array} i \nearrow k = \sum_k N_R^{ij} \bigcirc \nearrow k$$

$$= \sum_k N_k^{ij} d_k$$

$$d_X = \left[(F_x^{X, \bar{X}, X})_{1,1} \right]^{-1}$$

$$\begin{array}{c} X \quad \bar{X} \quad X \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ X \end{array} = (F_x^{X, \bar{X}, X})_{1,1} \begin{array}{c} X \quad \bar{X} \quad X \\ \diagdown \quad \diagup \\ \bigcirc \\ \diagup \quad \diagdown \\ X \end{array}$$

$$\bigcirc \nearrow X = (F_x^{X, \bar{X}, X})_{1,1} \bigcirc \nearrow X \bigcirc \nearrow X$$

$$d_X = (F_x^{X, \bar{X}, X})_{1,1} (d_X)^2$$

physical meaning of dx :

$$N_k^{ij} dk = di dj$$

$$(N^i)_k^j dk = di dj$$

$$\cancel{j} \left(\cancel{N^i} \right) \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} = di \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}^{\cancel{j}}$$

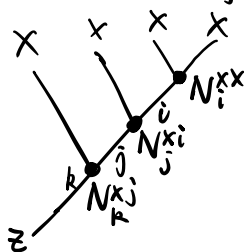
$\Rightarrow (d_1 \dots d_n)^T$ is the eigenvector of N^i with eigenvalue di .

$$N_k^{ij} \geq 0, di > 0$$

$\Rightarrow di$ is the largest eigenvalue of N^i .

Q: what is $\dim$ of $(X)^{\otimes n}$, $n \rightarrow \infty$?

A:



$$\dim X^{\otimes n} := \dim \left[\bigoplus_{\mathbb{Z}} \text{Hom}(X^{\otimes n}, \mathbb{Z}) \right]$$

$$\dim(X)^{\otimes n} = \sum_{i,j,k,\dots} N_i^{XX} N_j^{Xi} N_k^{Xj} \dots$$

$$= \sum_{i,j,k,\dots} (N^X)_i^X (N^X)_j^i (N^X)_k^j \dots$$

$$= \sum_{\mathbb{Z}} \left[(N^X)^n \right]_{X, \mathbb{Z}} \xrightarrow{n \rightarrow \infty} (dx)^n$$

$\Rightarrow \dim(X^{\otimes n}) \sim dx^n$ for $n \rightarrow \infty$

quantum dimension of X .

eg: $\dim \begin{pmatrix} \bullet & \bullet \\ \text{Majorana} & \text{Majorana} \end{pmatrix} = 2$

$$\Rightarrow \dim \begin{pmatrix} \bullet \\ \text{Majorana} \end{pmatrix} = \sqrt{2}$$

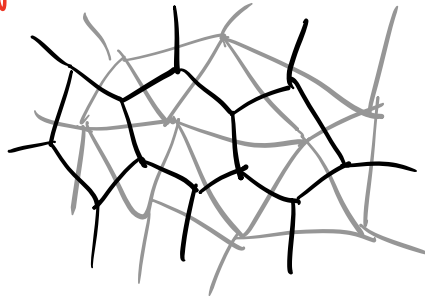
Finally, $\forall$ trivalent graph

$$\text{label} \begin{cases} \text{edge by } X \in \text{Obj}(\mathcal{C}) \\ \text{vertex by } \text{Hom}(X \otimes Y, Z) \text{ or } \text{Hom}(X, Y \otimes Z) \end{cases}$$

$\begin{array}{c} Y \\ \diagup \quad \diagdown \end{array}$
 $\begin{array}{c} \diagup \quad \diagdown \\ Z \end{array}$

$\updownarrow$ dual

triangulation of M_2



3.4. Examples.

Classification of fusion cat ?

Very hard!!! $\mathcal{C} = \text{Vec}_G \rightarrow$ classification of finite simple groups.
(already very hard!)

$$(1) \text{Obj}(\mathcal{C}) = \{1, e\} = \mathbb{Z}_2$$

$\vdots \quad |$

$$e \otimes e = 1, \quad e^* = e$$

associator

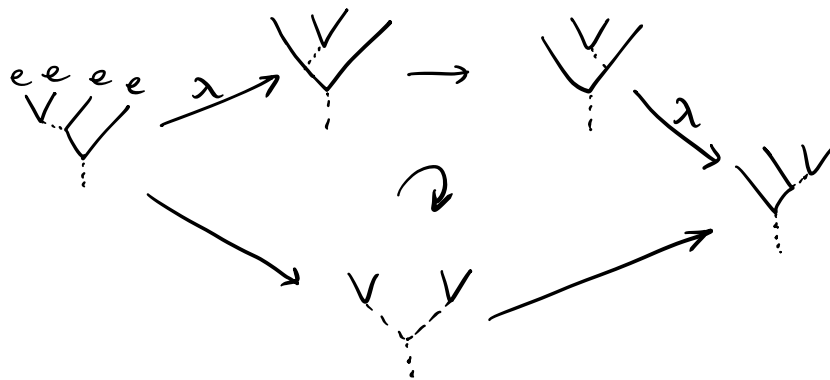
$a_{e,e,e} :$

$$\begin{array}{c} e \quad e \quad e \\ \diagdown \quad \diagup \quad \diagdown \\ \quad \quad \quad e \end{array} = \lambda \begin{array}{c} e \quad e \quad e \\ \diagup \quad \diagdown \quad \diagup \\ \quad \quad \quad e \end{array}$$

$a_{e,e,1} :$

$$\begin{array}{c} \diagdown \quad \diagup \\ \quad \quad \quad \vdots \end{array} = \begin{array}{c} \diagdown \quad \diagup \\ \quad \quad \quad \vdots \end{array}$$

pentagon:



$$\lambda^2 = 1 \Rightarrow \lambda = \pm 1$$

$$H^3(\mathbb{Z}_2, \text{val}) = \mathbb{Z}_2 \ni \nu_3 \begin{cases} \textcircled{1} \nu_3(a, b, c) = 1 \ (\forall a, b, c) : & \lambda = 1 \text{ toric code} \\ \textcircled{2} \nu_3(e, e, e) = -1 & : \lambda = -1 \text{ double semion.} \end{cases}$$

(2) Fibonacci .

$$\bullet \text{Obj}(\mathcal{C}) = \{1, \tau\}$$

$$\bullet \tau \otimes \tau = 1 \oplus \tau \rightsquigarrow \text{non-Abelian}$$

$$N_1^{\tau\tau} = N_{\tau}^{\tau\tau} = 1$$

$$\tau^{\otimes n} = F_{n-2} 1 \oplus F_{n-1} \tau \quad \text{where } F_n = 1, 1, 2, 3, 5, 8, \dots$$

is the Fibonacci number.

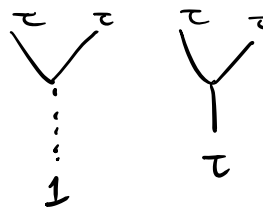
$$\begin{aligned} \tau \otimes (\tau^{\otimes n}) &= \tau \otimes (F_{n-2} 1 \oplus F_{n-1} \tau) = F_{n-2} \tau \oplus F_{n-1} (\tau \otimes \tau) \\ &= F_{n-2} \tau \oplus F_{n-1} (1 \oplus \tau) = F_{n-1} 1 \oplus (F_{n-2} + F_{n-1}) \tau \\ &= F_{n-1} 1 \oplus F_n \tau \end{aligned}$$

No class on 2021.10.20 and 2021.10.23 !

↓
Prof. Liang Kong's talk

• $\tau \otimes \tau = 1 \oplus \tau$

$\Rightarrow \text{circle with } \tau = \text{circle with } 1 + \text{circle with } \tau$



$\Rightarrow (d_\tau)^2 = 1 + d_\tau$

$\Rightarrow d_\tau = \frac{1+\sqrt{5}}{2} = \phi$ Golden ratio

• F symbol.

$= \sum_f (F_d^{abc})_{e,f}$

① $\tau \otimes \tau \otimes \tau \rightarrow 1$

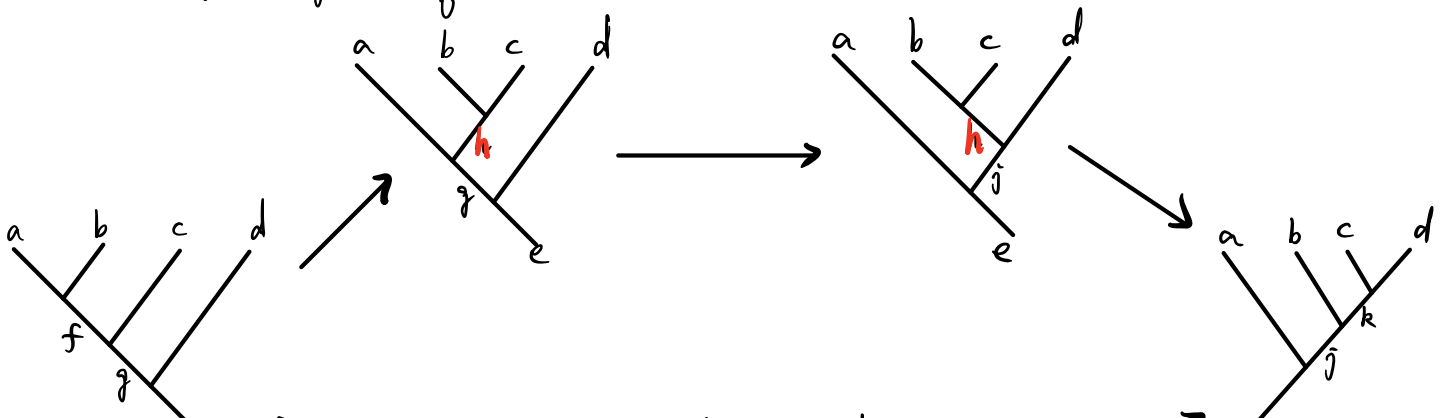
$= F_1^{\tau\tau\tau}$ $F_1^{\tau\tau\tau} = 1$

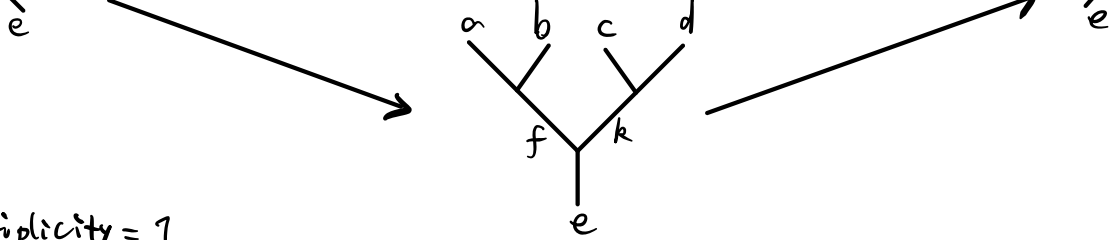
② $\tau \otimes \tau \otimes \tau \rightarrow \tau$

$= \begin{pmatrix} (F_\tau^{\tau\tau\tau})_{1,1} & (F_\tau^{\tau\tau\tau})_{1,\tau} \\ (F_\tau^{\tau\tau\tau})_{\tau,1} & (F_\tau^{\tau\tau\tau})_{\tau,\tau} \end{pmatrix} \begin{pmatrix} \text{Tree diagram} \\ \text{Tree diagram} \end{pmatrix}$

$\begin{pmatrix} \phi^{-1} & \sqrt{\phi^{-1}} \\ \sqrt{-\phi^{-1}} & -\phi^{-1} \end{pmatrix}$

• pentagon eq:





multiplicity = 1

$$\sum_h (F_g^{abc})_{f,h} (F_e^{ahd})_{g,j} (F_{\hat{j}}^{bcd})_{h,k}$$

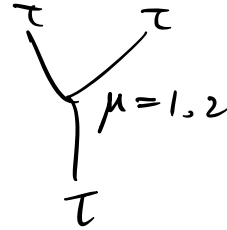
$$= (F_e^{fcd})_{g,k} (F_e^{abk})_{f,j}$$



$$\mu \in \text{Hom}(a \otimes b, c) = \mathbb{C}^{\text{(multiplicity)}}$$

If $\tau \otimes \tau = 1 \oplus \tau \oplus \tau$, then

$\uparrow$ $\uparrow$
 $\mu=1$ $\mu=2$



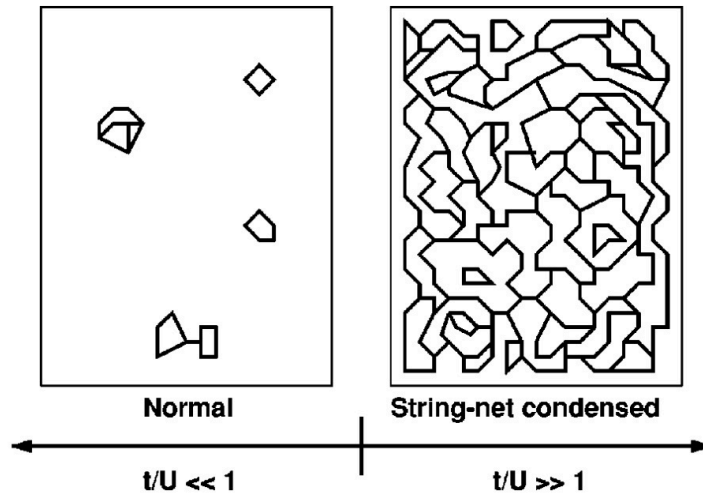
3.5. Levin-Wen model

PRL 71, 045110 (2005)

- string-net condensation
 ↓
 trivalent graph
 with fusion category labels

$$\langle b \rangle \neq 0 \quad |\psi_{GS}\rangle = \sum_n \#(b^+)^n |0\rangle$$

$$\Sigma (\text{all possible string-net conf.})$$



Example: $H = - \sum_s \begin{array}{c} \times \\ \diagup \quad \diagdown \\ s \end{array} - t \sum_p \begin{array}{c} \square \\ \diagup \quad \diagdown \\ p \end{array} - U \sum_{\text{links}} \frac{X}{\text{links}}$

..... $X = +1$
 ————— $X = -1$

new string tension term

① $t/U \ll 1, U \rightarrow +\infty$: $X = +1$ for all links

$$|\psi_{GS}\rangle = \bigotimes_{\text{link } \ell} |X=+1\rangle_{\ell} \rightarrow \text{product state.}$$

② $t/U \gg 1, U \rightarrow 0$: TC,

$$|\psi\rangle = \Sigma (\text{all closed loops}) \rightarrow \text{string-net cond.}$$

- string-net configuration:

color / label trivalent graph (V, E) by the fusion category $\mathcal{C}$.

$$E \rightarrow \text{obj}(\mathcal{C})$$

$$V \rightarrow \text{Hom}_{\mathcal{C}}(i \otimes j, k)$$

(1) string type:

$\uparrow^i$ simple object $i = 0, 1, \dots, N$ (N finite)

(2) branching rule:



$\mu \in \text{Hom}(i \otimes j, k)$ assume $\begin{cases} 0, & \text{if } i, j \text{ can not fuse into } k. \\ 1, & \dots \text{ can } \dots k \end{cases}$
 multiplicity = 1

(3) string orientation:

$$\uparrow^i = \downarrow^{i^*}$$

• Ground state wave function

$$|\Psi_{GS}\rangle = \sum_{\substack{\text{string-net} \\ \text{conf. } X}} \Phi(X) |X\rangle$$

$\{\Phi(X)\}$ are related to each other under local move $X \rightarrow X'$:

$$\Phi(\uparrow^i) = \Phi(\curvearrowright^i)$$

invariant under ambient isotopy

$$\Phi(\bigcirc^i) = d_i \Phi(\text{empty})$$

$$\Phi(\bigcirc_{ij}^i) = s_{ij} \Phi(\uparrow^i)$$

} scale invariance

$$\Phi\left(\begin{array}{c} i \quad j \quad k \\ \diagdown \quad \diagup \\ m \quad n \\ \diagup \quad \diagdown \\ \quad \quad \quad l \end{array}\right) = \sum_n (F_{\ell}^{ijk})_{mn} \Phi\left(\begin{array}{c} i \quad j \quad k \\ \diagdown \quad \diagup \\ \quad \quad n \\ \diagup \quad \diagdown \\ \quad \quad \quad \ell \end{array}\right)$$

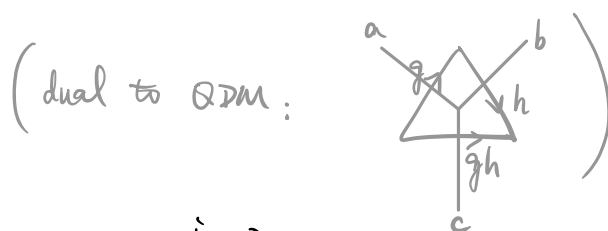
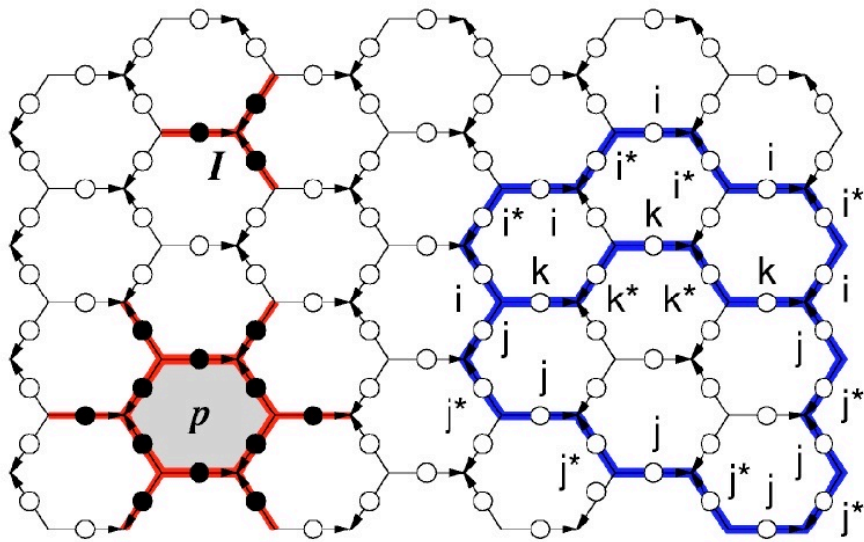
consistency condition: pentagon equation, rigid structure, ...

• Exactly solvable commuting-projector Hamiltonian

$$H = - \sum_s A_s - \sum_p B_p$$

$\downarrow$
constraint
on conf.

$\downarrow$
"gauge transf"



3-body interaction $\leftarrow A_s \left| \begin{smallmatrix} i & j \\ s & k \end{smallmatrix} \right\rangle = \delta_{ijk} \left| \begin{smallmatrix} i & j \\ s & k \end{smallmatrix} \right\rangle$, $\delta_{ijk} = \begin{cases} 1, & N_{k,j}^i \neq 0 \\ 0, & N_{k,j}^i = 0 \end{cases}$

$$B_p = \frac{1}{D^2} \sum_{s \in \text{obj}(e)} \underset{\substack{\uparrow \\ q \text{ dim}}}{d_s} \cdot B_p^s, \quad D^2 := \sum_s d_s^2 \text{ total } q \text{ dim.}$$

$$B_p^s \left| \text{hexagon} \right\rangle = \left| \text{hexagon with } s \text{ in center} \right\rangle = \sum \# \left| \text{hexagon with } s \text{ in center and } \text{red circle} \right\rangle$$

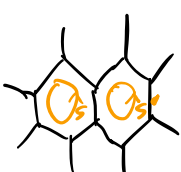
similar $\leftarrow A_s = \frac{1}{|e|} \sum_g A_s^g$

$$= \sum \# \left| \text{hexagon with } s \text{ in center and } \text{red circle} \right\rangle = \dots = \left| \text{hexagon with } s \text{ in center and } \text{red circle} \right\rangle = \dots$$

$$= \sum \# \left| \text{hexagon with } s \text{ in center and } \text{red circle} \right\rangle = \sum \# \left| \text{hexagon with } s \text{ in center and } \text{red circle} \right\rangle$$

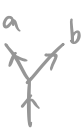
12-body interaction $\leftarrow B_p^s \left| \begin{smallmatrix} f & e \\ a & h & p & k \\ b & i & j & d \\ c \end{smallmatrix} \right\rangle = \sum F^{x6} \left| \begin{smallmatrix} f & e \\ a' & h' & p & k' \\ b & i & j' & d \\ c \end{smallmatrix} \right\rangle$

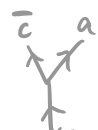
properties :

$$\begin{cases} [A_s, A_{s'}] = 0 \\ [B_p, B_{p'}] = 0 \\ [A_s, B_p] = 0 \\ A_s^2 = A_s \\ B_p^2 = B_p \end{cases}$$


$$B_p^2 = \left(\frac{1}{\mathcal{D}^2} \sum_s d_s \bigcirc_s \right)^2 = \frac{1}{\mathcal{D}^4} \sum_{ss'} d_s d_{s'} \bigcirc_{s'} \uparrow_s$$

$$= \frac{1}{\mathcal{D}^4} \sum_{ss'} \sum_{t\mu} d_s d_{s'} \bigcirc_{\mu}^{s' s} \uparrow_t = \frac{1}{\mathcal{D}^4} \sum_{ss'} \sum_{t\mu} d_s d_{s'} \bigcirc_{\mu}^{s' s} \uparrow_t$$





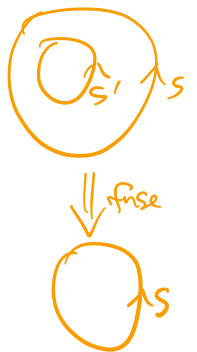
$$\text{Hom}(a \otimes b, c) \cong \text{Hom}(\bar{c} \otimes a, \bar{b})$$

$$N_c^{ab} = N_{\bar{b}}^{\bar{c}a}$$

$$= \frac{1}{\mathcal{D}^4} \sum_{ss't} d_s d_{s'} N_t^{s's} \bigcirc_t \uparrow_t$$

$$= \frac{1}{\mathcal{D}^4} \sum_{ss't} d_s d_{s'} N_{\bar{s}}^{\bar{t}s'} \bigcirc_t \uparrow_t$$

$$B_p = \sum_s d_s \bigcirc_s$$



$$d_s = \bigcirc_s \uparrow_s = \bigcirc_s \uparrow_s = d_{\bar{s}} \Rightarrow \frac{1}{\mathcal{D}^4} \sum_{s't} d_{s'} \left(\sum_{\bar{s}} N_{\bar{s}}^{\bar{t}s'} d_{\bar{s}} \right) \bigcirc_t \uparrow_t$$

$$d_i d_j = \sum_k N_k^{ij} d_k \Rightarrow \frac{1}{\mathcal{D}^4} \sum_{s't} d_{s'} (d_{\bar{s}} d_{s'}) \bigcirc_t \uparrow_t$$

$$= \frac{1}{\mathcal{D}^2} \left[\frac{1}{\mathcal{D}^2} \sum_{s'} (d_{s'})^2 \right] \sum_t d_t \bigcirc_t \uparrow_t$$

$$= \frac{1}{\mathcal{D}^2} \sum_t d_t \bigcirc_t \uparrow_t$$

$$= B_p$$

$\Rightarrow$ commuting-projector Hamiltonian

$$\Rightarrow \begin{cases} A_s |\Psi_{AS}\rangle = |\Psi_{AS}\rangle \\ B_p |\Psi_{AS}\rangle = |\Psi_{AS}\rangle \end{cases}$$